

AZIM PREMJI UNIVERSITY, BHOPAL

First Semester 2026-2027

Tutorial Sheet-5

Course No. MTH-113

Course title: Introduction to Quantitative Thinking

(1) Solve the equation to find value of x in each case.

(a) $9x + 2 = 5$

(b) $18x + 34x = 24$ (c) $7x + 18 = 16x - 35$

(d) $\frac{3x-4}{2} + \frac{5}{8} = \frac{24}{5} - \frac{x+2}{5}$ (e) $\frac{x-1}{4} = \frac{x-2}{3}$ (f) $18 - 4x = 12 - 32x$

(2) A wire is $(7x - 3)$ meters long. A length of $(3x - 4)$ meters is cut for use. Now, answer the following questions:

(a) How much wire is left?

(b) If this left out wire is used for making an equilateral triangle. What is the length of each side of the triangle so formed?

(3) One third of a number added to 6 is 17. Then, what is the square of this number?

(4) Thrice a number increased by 8 is 16. Find the number.

(5) Megha's age (in years) is 2 more than 5 times her daughter's age. If daughter's age is 5 years now, what is Megha's age?

(6) If $(2, 0)$ is a solution of the linear equation $2x + 3y = k$, then find the value of k .

(7) The linear equation $2x - 5y = 7$ has

(a) A unique solution

(b) Two solutions

(c) Infinitely many solutions

(d) No solution

(8) Any solution of the linear equation $2x + 0y + 9 = 0$ in two variables is of the form

(a) $(-\frac{9}{2}, m)$ (b) $(n, -\frac{9}{2})$

(c) $(0, -\frac{9}{2})$ (d) $(-9, 0)$

(9) Do the following pair of linear equations have no solution? Justify your answer.

(a) $2x + 4y = 3$

$12y + 6x = 6$

(b) $x = 2y$

$y = 2x$

(c) $3x + y - 3 = 0$

$2x + \frac{2}{3}y = 2$

(10) Solve the following system of linear equations.

(a)

$$2x + 3y = 12, x - y = 1$$

(b)

$$4x + 7y = 43, 3x - 2y = 4$$

(c)

$$5x - 3y = 7, 2x + 4y = 26$$

- (11) For which value(s) of λ , do the pair of linear equations

$$\begin{aligned}\lambda x + y &= \lambda^2 \\ x + \lambda y &= 1\end{aligned}$$

have

- (a) no solution?
 - (b) infinitely many solutions?
 - (c) a unique solution?
- (12) For which value(s) of k will the pair of equations
- $$\begin{aligned}kx + 3y &= k - 3 \\ 12x + ky &= k\end{aligned}$$
- have no solution?
- (13) For which values of a and b , will the following pair of linear equations have infinitely many solutions?
- $$x + 2y = 1, \quad (a - b)x + (a + b)y = a + b - 2$$
- (14) At a theater, the cost of an adult ticket is different from that of a child ticket. The cost of 3 adult tickets and 2 child tickets is Rs 460. The cost of 2 adult tickets and 4 child tickets is Rs 520. Find the cost of one adult ticket and one child ticket.
- (15) The sum of two numbers is 50 and their difference is 8. Find the two numbers.
- (16) Vijay had some bananas, and he divided them into two lots A and B. He sold the first lot at the rate of Rs 2 for 3 bananas and the second lot at the rate of Re 1 per banana, and got a total of Rs 400. If he had sold the first lot at the rate of Re 1 per banana, and the second lot at the rate of Rs 4 for 5 bananas, his total collection would have been Rs 460. Find the total number of bananas he had.
